# Type MS · IPv4 and IPv6: sizes, formats and why IPv6 exists

## Answer 1 of 3 · Question 4 part e · 3 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part e in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 4(a) | Award up to **two** marks for a linked explanation such as:<br><br>• 'What-if' questions can be asked by changing variables (1) to make predictions / see trends (1)<br><br>• The simulation can be rerun with different values (1) to see how the outcomes will change (1)<br><br>• The simulation can be run faster than real life (1) so multiple generations of animal results can be seen / long-term trends seen (1)<br><br>• The simulation can be safer (1) because of the danger of animal interaction (1)<br><br>• The simulation would be more convenient / saves time (1) because observers would have to be present outside for a long time (1) | Looking for a relevant statement and a linked extension/expansion.<br><br>Do not award two independent statements that are not linked.<br><br>Points could be reversed such as: you can make predictions because variables can be changed to see how the outcome will differ. | 2 |
| 4(b) | **The only correct answer is C**<br><br>***A** is not correct because it is not a linear scenario*<br>***B** is not correct because parallel models do not have to communicate and coordinate their actions*<br>***D** is not correct because there is not an explicit input with an expected output* | | 1 |

# Type MS · IPv4 and IPv6: sizes, formats and why IPv6 exists

## Answer 1 of 3 · Question 4 part e · 3 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part e in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(c) | Award up to **two** marks for a linked description such as:<br>• The research facility could use Two/Multi Factor Authentication (2FA/MFA) (1) sending a secondary piece of data to the employee which they then enter (e.g. passcode / authenticator app / email) (1)<br>• Using a username and password (1) that is unique to each individual employee / that matches one stored in a database (1)<br>• The research facility could implement a security question (1) that only the employee would know the answer to (1) | Do not allow any biometric measure e.g. fingerprint - biometrics is given in the question.<br><br>Do not accept password on its own - this is insufficient to allow the user's identity to be authenticated | 2 |
| 4(d)(i) | Award **one** mark for any of the following:<br>• The manipulation of matter with a size from 1 to 100 nanometres / scale of $10^{-9}$ / at atomic/molecular level (1)<br>• The inclusion of nanoparticles into different materials (1)<br>• Facilitates miniaturisation at atomic/molecular **scale/level** (1) | Small/miniature/microscopic is not enough. Responses must relate to the scale at the nano/molecular/atomic level. | 1 |
| 4(d)(ii) | Award **one** mark for any of the following:<br>• Stronger materials e.g. scratch resistant coatings / self-cleaning glass (1)<br>• Miniaturisation of electronics e.g. transistors (1)<br>• Nanomedicines to deliver drugs (1) | Allow examples that reasonably map to the categories given. | 1 |



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| | | | |
|---|---|---|---|
| 4(e) | Award **one** mark for each of the following up to a maximum of **three** marks:<br>• It uses a total of 128 bits (1)<br>• There is a total of 32 hexadecimal character (1)<br>• Each hexadecimal character requires 4 bits (1)<br>• There are 8 blocks of 4 character / 8 x 4 blocks (1)<br>• Each block (of 4 character) requires 16 bits (1)<br>• The address uses 4 (bits per character) x 32 (characters) / 4 x 32 (1)<br>• Each of the 8 blocks uses 16 bits / 8 x 16 (1) | | 3 |
| 4(f) | Indicative content<br>• Rights and responsibilities<br>• Data Protection regarding personal data<br>• Surveillance<br>• Consent of the individual to having specified data held<br>• Accountability of those running / using such systems<br>• Ownership<br>• Data mining & analysis of work patterns and potential use/misuse<br>• Ethics versus the legality of implementing such systems<br>• Data misuse and protection against hacking and malware<br>• Requirement to keep data secure<br>• Physical access to areas of the site<br>• Monitoring employee movement to restricted areas / resources<br>• Algorithmic bias in facial recognition | | 6 |
| | | **Total for question 4** | **16** |



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| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1-2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured. |



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# Type MS · IPv4 and IPv6: sizes, formats and why IPv6 exists

## Answer 1 of 3 · Question 4 part e · 3 marks

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · IPv4 and IPv6: sizes, formats and why IPv6 exists

Answer 2 of 3 · Question 1 part d · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(a) | Any **two** from:<br>• (Share) access to the Internet/WWW/broadband connection (1)<br>• (Internal) communication (using email/instant messaging/calendars) (1)<br>• Share files/data (1)<br>• Share peripherals/printers / hardware (1)<br>• Increases the amount of storage (since network storage devices can be attached) (1)<br>• Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1)<br>• Centralised backup (1)<br>• Centralised security (1) | • Award examples if mapped to a bullet point<br>• Do not award the same bullet more than once<br>• Do not award for install updates (in the stem) | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) (i) | Any **two** from:<br>• ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1)<br>• The connection does not get worse the further you are from the router / more reliable (1)<br>• Connection does not get obstructed by walls, ceilings, and furniture (1)<br>• More secure (1) | Accept reverse arguments for disadvantages of wireless. | 2 |

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Answer 2 of 3 · Question 1 part d · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part d in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) (ii) | Any **two** from:<br>• Installation/maintenance is more complex (1)<br>• Devices need to be physically connected (1)<br>• Less portable / limited by length of cable (1)<br>• Limited number of devices can be connected (1)<br>• Some digital devices cannot use a wired connection (1)<br>• Trip hazard (1) | Accept reverse arguments for advantages of wireless. | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c) | Award **two** marks for a linked description that addresses the individual mark points:<br><br>The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1)<br><br>The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|



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Answer 2 of 3 · Question 1 part d · 2 marks

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| | | | |
|---|---|---|---|
| 1(d) | Award **two** marks for a linked explanation such as:<br><br>Running out of IPv4 addresses/the number of (internet) connected devices has grown (1)<br><br>IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e)(i) | Star | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) (ii) | Any **one** from:<br><br>• easy to connect/remove new nodes (1)<br>• failure of one node/link does not affect the rest of the network (1)<br>• easy to detect the failure of one node/link (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) (iii) | Any **one** from:<br><br>• if central switch/hub fails then the whole network fails (1)<br>• performance and number of nodes that can be added depends on capacity of central switch/hub (1)<br>• can require more cable than some of the other topologies (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|



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Answer 2 of 3 · Question 1 part d · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(f)(i) | Any **two** from:<br>• SMTP (1)<br>• IMAP (1)<br>• POP/POP3 (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(f)(ii) | Any **two** from:<br>• passes the (reassembled) packets to the application layer (1)<br>• check if all packets have arrived (1)<br>• determine whether the contents are correct (1)<br>• requests resending of lost or damaged packets (1)<br>• reassembles packets in correct order/ into an email (1) | Do not accept receives from network layer (stem) | 2 |



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## Type MS · IPv4 and IPv6: sizes, formats and why IPv6 exists

### Answer 2 of 3 · Question 1 part d · 2 marks

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

# Type MS · IPv4 and IPv6: sizes, formats and why IPv6 exists

## Answer 3 of 3 · Question 4 part b · 2 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a) | **The only correct answer is C**<br><br>***A*** *is not correct because* ***mebibits*** *is a measure of storage capacity*<br><br>***B*** *is not correct because* ***mebibytes*** *is a measure of storage capacity*<br><br>***D*** *is not correct because transmission speeds are measured only in bits per second, not* ***bytes per second*** | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(b) | Award **one** mark for each correct cell:<br><br>**Type** / **Example** / **Number of bits**<br>IPv4 / 192.169.0.3 / 32 (1)<br>IPv6 / 1050:a500:00c0:0440:0006:0300:700d:436f / 128 (1) | | 2 |

# Type MS · IPv4 and IPv6: sizes, formats and why IPv6 exists

## Answer 3 of 3 · Question 4 part b · 2 marks

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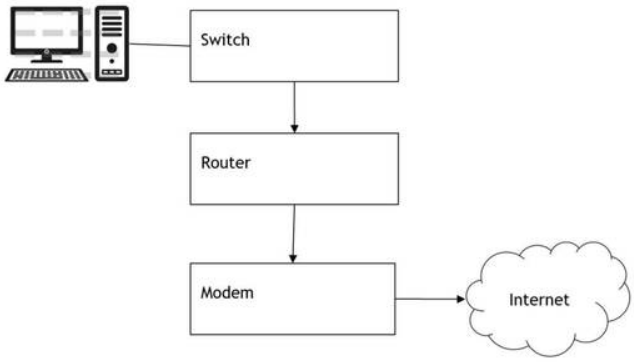
| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(i) | Award up to **two** marks for a linked description, such as:<br>• In a wired network packets travel over (copper/fibre-optic) cables (1), whereas in a wireless network packets travel over the air (1) | Allow radio signals, infra-red, microwave for air (must be more than just waves)<br><br>Allow twisted pair, CAT for copper.<br>Allow electrical/light signals (not just electricity) | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>1. A protocol (suite/group/family) (1)<br>2. Used on a wired network / wired connection (1)<br>3. Defines physical parts/type of cable(twisted pair, CAT6)/type of connector (1)<br>4. Defines how packets are checked for errors (1)<br>5. Defines the speed of transmission (1)<br>6. Operates at the link layer of the TCP/IP stack (1) | | 2 |

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### Answer 3 of 3 · Question 4 part b · 2 marks

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(d) | Award **three** marks for all of switch, router and modem in the correct order as in diagram<br><br>Award **two** marks for all of switch, router and modem in any order<br><br>Award **one** mark for any of switch or router or modem in any box<br><br>  <br> | | 3 |

# Type MS · IPv4 and IPv6: sizes, formats and why IPv6 exists

## Answer 3 of 3 · Question 4 part b · 2 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(e) | Award up to **two** marks for a linked description, such as:<br><br>Validation is used to check that the data/password entered by a user is fit for processing/meets specified requirements (1), whereas authentication is used to verify the identity of the user/prove that users are who they say they are (1) | For validation 'specified requirements' a good description of the check can be given e.g. presence check, length check, format check | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|